Partial replacement of soybean meal with fermented corn husk enhances growth, digestive enzymes, immune response, and growth-, immune-, and antioxidant-related gene expression in Nile tilapia (*Oreochromis niloticus*) reared in biofloc system

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Highlights

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Fermented corn husk (FCH) was evaluated as a sustainable partial replacement for soybean meal in tilapia diets.

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FCH supplementation at 10 g kg^{-1} significantly improved growth performance and feed conversion ratio.

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Digestive enzyme activities (amylase, lipase, protease, trypsin) were enhanced in fish fed FCH-supplemented diets.

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FCH modulated immune and antioxidant gene expression in a tissue-specific manner (head kidney and intestine).

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The study supports circular bioeconomy by converting agro-waste into functional aquafeed ingredients.

Abstract

This study investigated the effects of dietary fermented corn husk (FCH) on growth performance, digestive enzyme activities, immune responses, and gene expression in Nile tilapia (*Oreochromis niloticus*). A total of 300 fingerlings (40.18 ± 0.18 g) were randomly distributed into 15 tanks (three replicates per treatment, 20 fish per tank) and fed diets containing 0 (control), 2.5, 5, 10, or 20 g kg^{-1} FCH for eight weeks under biofloc conditions. Results showed that fish fed 10 and 20 g kg^{-1} FCH had significantly greater final weight, weight gain, specific growth rate, and lower feed conversion ratio compared to the control. Activities of digestive enzymes including amylase, lipase, protease, and trypsin were significantly elevated, especially in the FCH10 and FCH20 groups. Mucosal and serum immune responses, such as skin mucus and serum lysozyme, peroxidase, and ACH50 activities were markedly improved in FCH-fed fish, with optimal responses observed at 10 g kg^{-1}. Expression of immune-related (*il-1β, tnf-α, mhcii-α, nfkb*), antioxidant-related (*hsp70, gpx, nrf2*), and appetite-regulating genes (*ghrelin, npy-α,* and *galanin*) was significantly upregulated in the FCH5 and FCH10 groups. Additionally, *ef1-α*, a marker for protein synthesis, was significantly enhanced in the FCH5 and FCH10 groups, indicating improved anabolic activity, but declined at 20 g kg^{-1}. These findings highlight FCH, particularly at 10 g kg^{-1}, as a sustainable feed additive that improves growth, digestive efficiency, immunity, and molecular health in Nile tilapia.

Introduction

Aquaculture plays a vital role in global food production and sustainability efforts (Woźniacka et al., 2025; Xia, 2025). Over recent decades, advancements in aquaculture strategies have contributed to meeting the rising demand for seafood (Yang et al., 2025a). Nile tilapia (*Oreochromis niloticus*) is among the most widely farmed freshwater fish due to its rapid growth, adaptability to diverse environments and diets, and resilience to challenging conditions (Yang et al., 2025b). Over the past eight years, Nile tilapia production has risen by 70 %, increasing its per capita consumption (Al-Sisi et al., 2024). However, the high cost of feed remains a major constraint in expanding tilapia farming, as it accounts for over 75 % of total production expenses in intensive aquaculture systems (Musa et al., 2025; Rossignoli et al., 2023). The rising prices of essential protein sources, such as fishmeal and soybean meal, pose challenges for fish farmers due to market fluctuations and limited supply (Cantillo and Deshpande, 2025; Fantatto et al., 2024). To address this challenge, researchers are investigating alternative feed ingredients that offer cost-effectiveness, widespread availability, and avoid competition with resources intended for carnivorous aquaculture species and human consumption. (Iheanacho et al., 2025; Yadav et al., 2025). Sustainable aquaculture solutions will require innovative feed technologies, efficient resource utilization, and environmentally friendly production systems without compromising fish performance (Khan et al., 2024; Wani et al., 2025).

Many countries face issues related to agricultural waste management, including residues from crop production, which are generally managed as landfill, even though they might be repurposed as feed ingredients to enhance global food security and reduce carbon footprint when they treated as waste (Agya, 2025; Javed et al., 2025). Effective recycling strategies are essential to mitigate environmental pollution and reduce feed costs (Hajam et al., 2023; Yang et al., 2023). Maize (*Zea mays* L.), commonly known as corn, is an annual herbaceous crop belonging to the Poaceae family and the Panicoideae subfamily (Frosi et al., 2024). According to FAO data, maize ranks among the most widely consumed cereal crops globally, with China consistently cultivating it on over 40 million hectares of land (Cai et al., 2024; Frosi et al., 2024). Corn husk, an agricultural by-product of maize cultivation, is considered an environmentally friendly waste material. Annually, an estimated 20 million tons are produced, offering significant potential as a source of nanocellulose due to its high cellulose content (30–50 %) (Chawla et al., 2023; Mehmood et al., 2024). The main structural components of corn husk include cellulose, lignin, and various polysaccharides (Cai et al., 2024). As a significant by-product of the corn processing industry, corn husk is commonly utilized in animal feed and fermentation applications. Additionally, it exhibits pharmacological properties, including lipid-lowering effects, blood pressure regulation, and protection against intestinal disorders (He et al., 2024; Jing et al., 2022; Song et al., 2023). Earlier studies on corn husk have primarily concentrated on isolating dietary fibers and polysaccharides (Boukid, 2023). However, numerous bioactive compounds, including carotenoids and phenolics, remain underexplored and underutilized. As a major residue of maize cultivation, corn husk is abundant in phenolic compounds, particularly hydroxycinnamic acids like ferulic and ρ-coumaric acids, which contribute significantly to its antioxidant capacity (Galeana-López et al., 2021a, Galeana-López et al., 2021b). However, one challenge with corn husk is its high content of non-starch polysaccharides (Song et al., 2023), which can increase gut viscosity and hinder enzyme diffusion, reducing nutrient absorption (Karim et al., 2024). To overcome this limitation, solid-state fermentation (SSF) is a promising green technology that enhances the nutritional quality of agricultural byproducts. It has been reported that fermentation of corn husk fermentation process using starter of *Aspergilus niger* 5 % to 2 weeks can increase the production of VFA-NH_3 (Tampoebolon et al., 2019). Fermentation products have been shown to enhance fish growth rates, immunity, digestibility, and modulate gut microbiota (Mohammady et al., 2024; Phinyo et al., 2025; Qu et al., 2025; Roslan et al., 2025; Shimul et al., 2024; Wang et al., 2024a). Moreover, the fermentation process hydrolyzes fibers and proteins, releasing bioactive compounds that may provide additional health benefits (Nemes et al., 2025; Ozkan et al., 2025; Wu et al., 2025).

Biofloc technology (BFT) is a sustainable and eco-friendly aquaculture approach that transforms waste nutrients, particularly nitrogen, into microbial biomass, which can be utilized by cultured aquatic species or incorporated into feed formulations (Zhang et al., 2025). BFT has been shown to improve water quality, boost immune responses, enhance growth performance, and support disease resistance in aquaculture species (Aboseif et al., 2025; Behera et al., 2025; Bhadra et al., 2024). While some studies have evaluated the use of corn husk and their derivative as a feed additive in Nile tilapia (Galeana-López et al., 2021a, Galeana-López et al., 2021b), the effects of fermented corn husk in fish diets remains underexplored. Therefore, this study aimed to evaluate the effects of fermented corn husk on growth performance, digestive enzyme, immune response, and gene expression of Nile tilapia (*Oreochromis niloticus*) reared in BFT.

Sample preparation

Corn husks were collected, oven-dried at 60 °C for 48 h, ground, and sieved through a 100-mesh screen. The resulting powder was stored at 4 °C until use.

Pretreatment process

Ten grams of corn husk powder were mixed with 0.5 % NaOH and subjected to high-pressure processing at 50 MPa for 5 or 20 min. Samples were rinsed with deionized water, oven-dried at 80 °C for 24 h, and stored at 4 °C for analysis (Sahare et al., 2012).

Enzymatic hydrolysis

High-pressure pretreated corn husk (4 g) was mixed with 40 mL of 0.1 M citrate buffer (pH 4.8)

Growth performance

The initial body weight (BWi) of Nile tilapia did not differ significantly among treatments, ranging narrowly from 40.12 to 40.25 g (Table 5). After 8 weeks of feeding, final body weight (BWf) increased progressively with higher inclusion levels of fermented corn husk (FCH), reaching the highest value in the FCH20 group (222.58 ± 3.30 g). This increase followed a significant linear trend ($P < 0.05$, $R^2 = 0.97$), while the quadratic trend was not significant. A similar pattern was observed for

Discussion

Optimizing feed formulations using agricultural by-products is a sustainable strategy to enhance aquaculture productivity while reducing feed costs and environmental impact. In this study, fermented corn husk (FCH) was evaluated as a dietary ingredient for Nile tilapia over an 8-week feeding trial in biofloc technology (BFT). The findings revealed that replacement of soybean meal with FCH (FCH10 and FCH20) significantly improved growth performance and feed utilization without affecting fish

Conclusion

This study demonstrates that partial replacement of soybean meal with fermented corn husk (FCH) in Nile tilapia diets under a biofloc system can significantly enhance growth performance, digestive enzyme activity, immune response, and gene expression related to growth, immunity, and antioxidant defense. The optimal inclusion level was identified as 10 g kg^{-1}, at which fish exhibited the most consistent physiological and molecular improvements. Moreover, tissue-specific gene expression analysis

Animal ethics

All animal experiments comply with AAALAC guide-lines approval by Chiang Mai University Committee (Approval No.: AG03004/2567).

CRediT authorship contribution statement

Hien Van Doan: Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Supreya Wannavijit:** Formal analysis, Data curation. **Khambou Tayyamath:** Investigation, Data curation. **Tran Thi Diem Quynh:** Formal analysis, Data curation. **Md Afsar Ahmed Sumon:** Formal analysis, Data curation. **Nguyen Vu Linh:** Data curation. **Phisit Seesuriyachan:** Resources. **Yuthana Phimolsiripol:** Writing – review & editing. **Maria Ángeles Esteban:** Writing – review &

Declaration of competing interest

The authors whose names are listed immediately below certify that they have NO affiliations with or involvement in any organization or entity with any financial interest (such as honoraria; educational grants; participation in speakers' bureaus; membership, employment, consultancies, stock ownership, or other equity interest; and expert testimony or patent-licensing arrangements), or non-financial interest (such as personal or professional relationships, affiliations, knowledge or beliefs) in

Acknowledgments

The authors would like to thank Chiang Mai University and the National Research Council of Thailand for their support and assistance with this study. This study was partially supported by Chiang Mai University.